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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BVOC(HONOURS)REGULAR EXAMINATION NOVEMBER 2025
FIRST SEMESTER
SKILL DEVELOPMENT COMPONENTS(SDC)-MG1SDCBAT100-
FUNDAMENTALS OF HORTICULTURE AND NURSERY MANAGEMENT
(2025 ADMISSION ONWARDS)

PART A

1. Mention any two roles of horticulture in environmental improvement.
- Horticultural crops especially trees and perennial plantations,help in soil and water conservation
- It promotes ecological balance

2. What are the disadvantages of sexual propagation
- Seedling trees are not uniform in their growth
- Take more time to come to bearing

3. Write a short note on tip layering.
In tip layering,rooting takes place near the tip of the current seasons shoot which is bent to the ground.
Eg;Blueberry

4. List components of nursery
Progeny orchard, propagating structures-hot frame,cold frame,lath house,net house,mist,greenhouse.

5. The term horticulture is derived from the latin word:Hortus meaning…….and cultura meaning….
Hortus-garden cultura-cultivation

6. What is open centre system of training?
Horticulture training method where the main stem is cut back to encourage the growth of 3-5 main scaffold branches that radiate from the top of the trunk.

7. Briefly explain water management
Water management in an orchard means the efficient and timely application of water to fruit trees to meet their growth and yield requirements without damage.

PART B

8. Describe hexagonal system of planting.Discuss the advantages and disadvantages
In hexagonal system,the trees planted in the corners of equilateral triangles. Six trees thus form a hexagon with another tree at its center. It accomodates 15% more plants.
Advantages
- Accomodates 15% more plants than the square system.
- Plants are planted at the corners of equilateral triangle
- This requires fertile land

Disadvantages

- Lay out is difficult and cumbersome

9. Write a short note on propagation by stem cutting

Stem cuttings

- **Hardwood cuttings**-these cuttings are made from the past seasons growth or wood that has matured and lignified are known as hardwood cuttings. Propagation by hardwood cuttings is simple and cheapest method of multiplication. Eg;grape,pomegranate
- **Semihardwood cuttings**-mostly used for evergree fruit plants. The available shoots during june-july have not attained the full maturity and are 5-9 months old. Eg;mango,guava
- **Softwood cuttings**-cuttings are prepared from the soft succulent plants which are 4-6 months old. Eg; nerium,croton
- Herbaceous stem cuttings-cuttings is taken from succulent herbaceous greenhouse plants.Eg;cactus,chrysanthemum

10. Write a short note on maintenance of nursery

Maintenanance of nursey is the systematic raising of healthy and vigorous planting material for agriculture,horticulture,forestry and landscaping.

Site selection-a nursery should be located on well drained soil with good sunlight, easy access to water,transportation etc.

Soil preparation-the nursery soil should must be fertile,loose and free from pests and diseases.

Seed and planting material-use of high quality,certified seeds and disease free planting material.

Irrigation and nutrition-regular watering and organic manures are applied to supply essential nutrients for growth.

Pest and disease management-regular monitoring,sanitation and safe plant protection measures maintain plant health.

11. Differentiate in detail between training and pruning in horticultural crops

Training

Training refers to the judicious removal of plant part / parts to develop proper shape of a plant capable of bearing a heavy crop load.

(i) **Open centre system** (Vase shaped) In this system the main stem is allowed to grow to a certain height and the leader is cut to encourage lateral scaffold from near the ground giving a vase shaped plant. This is common in peaches, apricots

(ii) **Central leader system** (closed centre) In this system the central axis of plant is allowed to grow unhindered permitting branches all around. This system is also known as closed centre system and common in use in apple, pear, mango and sapota.

(iii) **Modified leader system** This system is in between open centre and central leader system wherein central axis is allowed to grow unhindered upto 4—5 years and then the central stem is headed back and laterals are permitted .It is common in apple, pear, cherry, plum, and guava

Pruning

Pruning is the removal of unwanted, surplus annual growth; dead, dried and diseased wood of the plants is called Pruning.

i. **Thinning out**:When a shoot is entirely removed from the point of its origin and no re-growth is allowed to occur from the cut ends.

ii. **Heading back**:When the terminal portion of branch/shoot is removed and it encourages lateral growth from the remaining shoot.
iii. **Notching**:Removal of wedge shaped bark piece from above the bud to make a notch is called as notching.
iv. **Nicking**:Removal of wedge shaped bark piece below the bud to make a notch is called as notching.

12. Write a note on precooling and its importance in extending shelf life of horticultural produce.

- Precooling is the rapid removal of field heat from freshly harvested horticultural produce before storage,packaging or transport.
- It slow down respiration and metabolic activities

Methods of precooling

- Forced air cooling
- Hydrocooling
- Vaccum cooling
- Evaporative cooling

Importance

- Reduces field heat
- Slow down respiration and ripening
- Maintains freshness
- Reduces microbial growth
- Improves quality

13. Briefly describe about lath house,net house and mist house.

Lath house-it is a valuable asset in raising seedling,rooted cuttings and young plants prior to setting them out in the garden. Lath house will modify the environment in which they are growing by offering protection from hot,drying summer winds,reducing the intensity of sunlight,lowering temperature and maintaining higher humidity.

Mist house-it is a technique for minimising plant moisture loss by controlled periodic wetting of the foliage of cuttings which are being rooted. It is helpful for rooting leafy cuttings.

Net house-net houses are widely used as propagation structures in tropical areas,where artificial heating is not required and artificial cooling is expensive. Net houses may be constructed with roofs covered with glass or plastic film and its sides are covered with wire net. It maintains ideal temperature for germination of seeds.

14. What is a nursery?Write a short note on the importance of nursery

Nursery is an area in which new saplings are raised and nourished until they are ready for sale or transplanting at a permanent place in a field.

Importance of nursery

- It is possible to grow and maintain a large number of plants per unit area.
- Small and expensive hybrid seeds can be raised more effectively due to better care and management.
- Germination percentage increases.
- Manipulation of growing conditions for plants becomes easy
- Better and uniform crop growth
- Off season sowing of seeds becomes possible
- Management of insect pests,diseases and weeds is easy in a nursery

PART C

15.Discuss in detail about propagation by budding

The plant that grows after union of the stock and the bud is known as budding.

- **T-budding**-the cuts given on the stock are of the shape of the letter T. It is widely used for propagating fruit trees and many ornamental plants. Eg;rose and citrus
- **Inverted T budding**-inverted T budding procedure is same as that of T budding except the horizontal cut on the stock is made at the bottom of the vertical cut rather than at the top.
- **Patch budding**-a regular patch of bark is completely removed from the stock plant and is replaced with a patch of bark of the same size containing a bud from the desired mother plant.Eg;walnut
- **Ring budding**-in the root stock,two transverse cut 1.5 cm apart are made and these are connected with a vertical cut and a ring of bark is removed.
- **Flute budding**-in flute budding,the patch of bark is removed from the stock is such a way that it completely encircles the stock except with a narrow bark connection between the upper and lower cuts on the stock.
- **Forkert budding**-the stock is prepared by giving two vertical cuts and a transverse cuts above the vertical cuts to join them. The bark is removed carefully along the cuts,so the flap of bark hangs down. The scion is then slipped into the exposed portion of the stock and the flap is drawn over the inserted bud patch. It is then tied with a wrapping material.
- **Chip budding**-a chip is removed from the budwood, ensuring that the bud is in the middle of the chip. The chip bud is then inserted in the stock in such a way that cambium of the bud chip should have direct contact with the cambium of the stock.

16.Write a short note on selection of location and site, nature of soil and climate in orchard planning.

Proper selection of sites is the basic need for growing crops. As the phenotypic characteristics of plants are affected by the combining effect of the genotype and the environment the grower must select the crop to grow according to the particular climatic conditions. The selection of the site must be based on the following parameters:

1. The location of orchards should be in well-established fruit growing regions as one could get the benefits from the experienced growers, easy market facility, etc.

2. Proper transportation facilities must be there as horticultural produce is perishable (decay or is destroyed quickly) in nature.

3. The market area should be close to the orchard.

4. Suitable climate: The climate of the region where one wants to develop its orchard, should be specific to the crop grown. There should be well-distributed rainfall and availability of optimum sunlight with a less diurnal variation. The climate should not be more humid, which may allow the growth of diseases and pests. For example, in grapes, high humidity favours the growth of powdery mildew fungus.

5. Adequate supply of water throughout the year.

6. Suitability of soil, its fertility, soil depth, and nature of subsoil must be observed. Soil with a pH at neutral range, along with loose subsoil is the best for the root environment. There should not be fluctuating water table, which may hinder the physiological activity of the root.
7. Site must have proper drainage during the rainy season.
8. Irrigation water must ob of good quality
9. Land should be cheap and plentily available nearer to the site for future expansion of the orchard.
10. The grower must be aware of whether the orchard is a new venture or one already established.
11. Sufficient availability of manure at a cheap rate in the locality.
12. Site must be free from all kinds of natural disasters like floods, drought, heavy winds, frost, etc.
13. Availability of skilled labour plentily and at a cheap rate in the locality of the orchard.